

Aadarsha Gopala Reddy

+1 (740) 802-1776 | adurs2002@gmail.com | LinkedIn | GitHub | agreddy.com

EDUCATION

Washington University in St. Louis <i>M.S. Computer Science</i>	St. Louis, Missouri, USA August 2024 - May 2026
• Thesis: Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data (Advisor: Dr. Alvitta Ottley) • Coursework: Deep Learning, Computer Vision, Computational Biology, Data Manipulation & Management at Scale, Artificial Intelligence for Health, Systems Security.	

Ohio Wesleyan University <i>B.A. Computer Science & Data Analytics</i>	Delaware, Ohio, USA August 2020 - May 2023
• Minor: Economics	

SCHOLARLY WORK

Gopala Reddy, Aadarsha, “Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data” (2026). M.S. thesis, Washington University in St. Louis. <https://doi.org/10.7936/g95n-fv35>

RESEARCH EXPERIENCE

Washington University in St. Louis <i>Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data</i>	St. Louis, Missouri, USA August 2025 - May 2026
---	---

Master’s Thesis with the DRIVES Project (PI: Dr. Ganesh Babulal; Advisors: Dr. Alvitta Ottley, Dr. Nathan Jacobs)

- Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 participants, deriving vehicle-agnostic weekly behavioral features and integrating demographic covariates for CDR status prediction.
- Compared six CDR prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-participant-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).

Ohio Wesleyan University <i>Opinion Survey on Artificial Intelligence in the Workplace</i>	Delaware, Ohio, USA January 2023 - May 2023
--	---

Independent Study (Mentor: Dr. Nicholas Dietrich)

- Designed, fielded, and investigated a survey experiment on the impact of AI on 250 employees within each of four industries using survey data.
- Analyzed data using Tableau dashboards and R to identify trends and common opinions.
- Presented preliminary results on AI’s positive impact on employee work at the OWU Spring Student Symposium.

<i>Artificial Intelligence in Modern Board Games — Lost Cities AI</i>	May 2022 - July 2022
---	-----------------------------

Summer Science Research Program (Mentor: Dr. Sean McCulloch)

- Developed a digital version of the Lost Cities card game using Java.
- Implemented an intelligent agent that played the game against a human, winning 13 of 18 games in testing.

TEACHING & MENTORSHIP EXPERIENCE

Washington University in St. Louis <i>Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale</i>	St. Louis, Missouri, USA January 2026 - May 2026
---	--

- Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing strategies, and query optimization.
- Designed oral midterms as mock technical interviews, requiring students to reason through pipeline architecture decisions and troubleshoot system failures under real-world constraints.
- Mentored three capstone teams from raw open datasets to production-grade dashboards, providing weekly feedback that kept all teams on schedule through final presentations.
- Held weekly office hours to debug student pipeline implementations across Snowflake schema design, Airflow DAG failures, and Kafka consumer configuration.

<i>Engineering Workshop by AGES & Science Coach</i>	September 2024
• Delivered an introductory workshop on artificial intelligence to high school students, covering core AI concepts with hands-on demonstrations.	

- Designed and facilitated a hands-on databases workshop for beginner-to-intermediate developers, teaching through two complete projects rather than lectures.
- Guided participants through building a normalized SQLite database from the iTunes Search API and a full-stack Supabase Todo app with Row Level Security — covering relational databases, APIs, cloud Postgres, and access control.

Ohio Wesleyan University

Delaware, Ohio, USA

*Computer Science Lab Assistant***January 2022 - May 2023**

- Tutored approximately ten students for over 6 hours per week, assisting with homework and exam preparation for introductory, intermediate, and advanced computer science coursework.
- Tested and graded assignments of around 35 students in introductory and intermediate computer science classes.

PRESENTATIONS**Ohio Wesleyan University**

Delaware, Ohio, USA

*Spring Student Symposium***April 2023**

Artificial Intelligence Opinion Survey

*Patricia Belt Conrades Summer Science Research Symposium***September 2022**

Artificial Intelligence in Modern Board Games

GRANTS, FELLOWSHIPS, & AWARDS

Bauer Leaders Academy (BLA) Leadership Development Badge , Washington University in St. Louis	April 2026
Affiliated School Scholarship , Washington University in St. Louis	May 2024
Dean's List , Ohio Wesleyan University	May 2023
Mortar Board National College Senior Honor Society Membership , Ohio Wesleyan University	April 2023
Golden Bishop Award - WCSA Best New Member , Ohio Wesleyan University	April 2023
Dean's List , Ohio Wesleyan University	May 2022
Florence Leas Sophomore Prize , Ohio Wesleyan University	April 2022
First Place - 32nd Annual Spring Programming Contest , Denison University	March 2022
International Baccalaureate Scholarship , Ohio Wesleyan University	January 2020
Scholar , Next Genius Scholarship Foundation	January 2020

PROFESSIONAL SERVICE & LEADERSHIP**Washington University in St. Louis**

St. Louis, Missouri, USA

*Member, Graduate Student Affairs Advisory Board (GSAAB)***August 2025 - May 2026**

- Provided strategic feedback to the Vice Chancellor for Student Affairs on critical issues impacting graduate student life and experience.
- Engaged in critical dialogue with senior university leaders, including the Dean of Students, to advocate for improvements to the graduate student experience across all academic programs.
- Contributed to monthly discussions on student affairs policies and initiatives to enhance support systems and resources for the graduate student community.

*Graduate Student Member, Center for Career Engagement (CCE) Student Advisory Board***August 2025 - May 2026**

- Guided the strategic direction of the Center for Career Engagement to ensure services align with diverse needs of graduate students from all academic backgrounds and career interests.
- Provided insights on career engagement programs, workshops, and resources to enhance accessibility and effectiveness for the WashU graduate student community.
- Built community around career development initiatives and fostered collaboration between the CCE and graduate student body.

*Vice President of the Graduate-Professional Council (GPC) Chamber, Graduate and Professional Student Council (GPSC)***May 2025 - May 2026**

- Built and maintained committee membership rosters using ranked-choice preference allocation to balance workload and ensure efficient cross-committee coordination.
- Oversaw information architecture for the Council's documentation repository.
- Represented the graduate student body to university leadership and supported the GPC President in advancing strategic council priorities.
- Chaired the Constitution Committee (August 2025 - February 2026), coordinating stakeholder input to draft and ratify GPSC governing documents aligned with the Council's mission and university policies.

Treasurer, Umang (Indian Graduate Student Association) **June 2025 - October 2025**

- Managed club finances and organized cultural events and festivals, promoting cultural awareness and fostering community among Indian graduate students.

Treasurer, HackWashU **December 2024 - July 2025**

- Managed the club's finances, including budgeting and expense tracking, to ensure sustainable operations and successful event planning.
- Coordinated fundraising efforts and sponsorship outreach to support club activities and initiatives.
- Collaborated with club members to organize hackathons and coding events, fostering a culture of innovation and teamwork among participants.

Ohio Wesleyan University **Delaware, Ohio, USA**

Budget Committee Senator, Wesleyan Council on Student Affairs (WCSA) **August 2022 - May 2023**

- Proposed and voted on senate bills while overseeing allocation of an annual budget exceeding \$350,000 to campus individuals and organizations.
- Received the Golden Bishop Award for WCSA Best New Member for overall contribution to WCSA's mission and service on the Budget Committee.

Member, Mathematics, Computer Science & Data Analytics Student Board **August 2021 - May 2023**

- Analyzed faculty evaluations and synthesized student feedback on course offerings to recommend departmental improvements.

Member, The Neurds (Neuroscience Club) **August 2022 - May 2023**

- Presented on AI at Brain Fair 2023 during Brain Awareness Week, bridging neuroscience and technology.

Vice President of Finance, Campus Programming Board **January 2022 - December 2022**

- Managed annual budgets exceeding \$86,000 for campus-wide programming and secured \$10,000 in additional annual funding through negotiation with WCSA.
- Revised the organization's governing documents, codifying financial and operational practices.

PROJECTS

Washington University in St. Louis **St. Louis, Missouri, USA**

Multimodal Prediction of Alzheimer's Disease **August 2024 - December 2024**

- Developed a multimodal fusion approach for early detection of Alzheimer's Disease, combining MRI imaging and clinical/demographic data from the OASIS-1 dataset.
- Implemented CNNs using TensorFlow/Keras for brain MRI analysis alongside XGBoost and Random Forest models for tabular clinical data.
- Created a combined classifier integrating both modalities and documented the project in NeurIPS format with comprehensive performance evaluation metrics and visualizations.

Red-Blue Visual Auto Defender: Automated Visual Jailbreak Generation and Explainable Defenses **September 2025 - October 2025**

- Built an automated red-blue teaming pipeline for vision-language model (VLM) security with teammates Stuart Aldrich and Mohammad Rouie Miab, addressing visual prompt injection (VPI) attacks against VLM-based agents.
- Generated visual jailbreak attack images by overlaying malicious instructions onto benign images and evaluated them against a target VLM (Gemma-3-4b-it) using semantic response analysis.
- Auto-generated deterministic, OCR-based Python defenses for each successful attack, validated with accuracy, precision, and recall metrics in an iterative refinement loop.

- Implemented multiple deep reinforcement learning algorithms (DDQN, PPO, SAC) integrated with EnergyPlus simulations to optimize datacenter cooling systems.
- Achieved 35.8% energy efficiency improvement over baseline using DDQN, addressing the underserved small to mid-sized datacenter market.
- Created a comprehensive framework integrating building energy simulation with reinforcement learning for dynamic thermal management.

MITxSureStart FutureMakers Create-a-Thon Program

Remote

Parkinson's Disease AI Diagnosis Software

June 2022 - July 2022

- Built a dual-input pipeline combining audio spectrograms (custom CNN for vocal tremor detection) and spiral-drawing images (Random Forest classifier with augmentation for micrographia detection).
- Implemented a weighted ensemble combining both modalities into a single Parkinson's diagnosis score.
- Designed a Flask web interface for user submissions of audio and drawing samples for real-time risk assessment.

OTHER PROFESSIONAL EXPERIENCE

Washington University in St. Louis

St. Louis, Missouri, USA

Graduate Assistant, Taylor Family Center for Student Success

August 2024 - May 2026

- Planned and ran three recurring engagement programs for first-generation, limited-income (FLI) students — Welcome Dinner, First-Gen Week, and End of Week Unwind — coordinating logistics, promotion, and on-the-day operations.
- Authored FERPA-compliant process manuals for engagement data collection and evaluation, giving center staff a standardized, repeatable framework for measuring FLI student participation.
- Analyzed semester survey responses and event attendance data, producing reports that directly informed the following cycle's programming decisions and outreach priorities.
- Managed the center's weekly newsletter; represented the department at off-campus excursions and FLI student trips.

Crittero, Inc.

Remote

AI Engineer Intern

June 2025 - August 2025

- Generated labeled social media training data for seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Trained a TensorFlow/Keras recommendation model with persona-type, content-category, and temporal embeddings, plus a collaborative-filtering cold-start fallback for new users.
- Built a configurable Python CLI for reproducible model evaluation, enabling systematic comparison across hyperparameter settings, embedding dimensions, and fallback thresholds.

Lab714

Boca Raton, Florida, USA

Data Analyst Intern

June 2023 - May 2024

- Analyzed IoT environmental sensor data streams on AWS, building dashboards that surfaced chemical imbalances and equipment anomalies through automated threshold detection and trend analysis.
- Built AWS data ingestion pipelines to normalize and store continuous sensor telemetry from deployed devices, supporting real-time operational monitoring of deployed IoT fleets.
- Wrote SQL queries to aggregate historical sensor data by pool, date range, and chemical parameter, powering the trend analytics and anomaly detection shown in client dashboards.
- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Java, C++, Python, R, JavaScript/TypeScript, PHP, C#, SQL, Rust.

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, Pandas, Matplotlib, Seaborn, Node.js, Express.js, Vue.js, React.

Tools & Technologies: L^AT_EX, Git, Tableau, Power BI, MySQL, AWS, MongoDB, Snowflake, Apache Airflow, Spark, Kafka, Flink.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.